

A Technical Introduction to Flash attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Section 1

In general, there are 3 classes of language modelling tasks: "masked", "autoregressive", and "prefixLM". These classes are independent of a specific modeling architecture such as transformer, but they are often discussed in the context of transformer. In a masked task, one or more of the tokens is masked out, and the model would produce a probability distribution predicting what the masked-out tokens are based on the context. The loss function for the task is typically sum of log-perplexities for the masked-out tokens: $\text{Loss} = -\sum_{t \in \text{masked tokens}} \ln(\text{probability of } t \text{ conditional on its context})$ and the model is trained to minimize this loss function. The BERT series of models are trained for masked token prediction and another task. In an autoregressive task, the entire sequence is masked at first, and the model produces a probability distribution for the first token. Then the first token is revealed and the model predicts the second token, and so on. The loss function for the task is still typically the same. The GPT series of models are trained by autoregressive tasks. In a prefixLM task, the sequence is divided into two parts. The first part is presented as context, and the model predicts the first token of the second part. Then that would be revealed, and the model predicts the second token, and so on. The loss function for the task is still typically the same. The T5 series of models are trained by prefixLM tasks. Note that "masked" as in "masked language modelling" is not "masked" as in "masked attention", and "prefixLM" as in "prefix language modeling" is not "prefixLM" as in "prefix language model".

Section 2

$\text{MultiQueryAttention}(Q, K, V) = \text{Concat}_{i \in [n \text{ heads}]} (\text{Attention}(XW_i^Q, XW_i^K, XW_i^V)) W^O$

Section 3

restoring or repairing incomplete or corrupted text. For example, the input, "Thank you ~ me to your party ~ week", might generate the output, "Thank you for inviting me to your party last week". translation between natural

languages (machine translation) judging the pragmatic acceptability of natural language. For example, the following sentence might be judged "not acceptable", because even though it is syntactically well-formed, it is improbable in ordinary human usage: The course is jumping well. Note that while each of these tasks is trivial or obvious for human native speakers of the language (or languages), they have typically proved challenging for previous generations of machine learning architecture.

Section 4

Disaster response biological sequence analysis video understanding protein folding (such as AlphaFold) evaluating chess board positions. Using static evaluation alone (that is, with no Minimax search) transformer achieved an Elo of 2895, putting it at grandmaster level.

Section 5

A decoder consists of an embedding layer, followed by multiple decoder layers, followed by an un-embedding layer. Each decoder consists of three major components: a causally masked self-attention mechanism, a cross-attention mechanism, and a feed-forward neural network. The decoder functions in a similar fashion to the encoder, but an additional attention mechanism is inserted which instead draws relevant information from the encodings generated by the encoders. This mechanism can also be called the encoder-decoder attention. Like the first encoder, the first decoder takes positional information and embeddings of the output sequence as its input, rather than encodings. The transformer must not use the current or future output to predict an output, so the output sequence must be partially masked to prevent this reverse information flow. This allows for autoregressive text generation. For decoding, all-to-all attention is inappropriate, because a token cannot attend to tokens not yet generated. Thus, the self-attention module in the decoder is causally masked. In contrast, the cross-attention mechanism attends to the output vectors of the encoder, which is computed before the decoder starts decoding. Consequently, there is no need for masking in the cross-attention mechanism. Schematically, we have: $H' = \text{MaskedMultiheadAttention}(H, H, H) \text{DecoderLayer}(H) = \text{FFN}(\text{MultiheadAttention}(H', H^E, H^E))$ where H^E is the matrix with rows being the output vectors from the encoder. The last decoder is followed by a final un-embedding layer to produce the output probabilities over the vocabulary. Then, one of the tokens is sampled according to the probability, and the decoder can be run again to produce the next token, etc., autoregressively generating output text.

Section 6

The idea of encoder–decoder sequence transduction had been developed in the early 2010s; commonly cited as the originators that produced seq2seq are two concurrently published papers from 2014. A 380M-parameter model for machine translation uses two long short-term memories (LSTM). Its architecture consists of two parts. The encoder is an LSTM that takes in a sequence of tokens and turns it into a vector. The decoder is another LSTM that converts the vector into a sequence of tokens. Similarly, another 130M-parameter model used gated recurrent units (GRU) instead of LSTM. Later research showed that GRUs are neither better nor worse than LSTMs for seq2seq. These early seq2seq models had no attention mechanism, and the state vector is accessible only after the last word of the source text was processed. Although in theory such a vector retains the information about the whole original sentence, in practice the information is poorly preserved. This is because the input is processed sequentially by one recurrent network into a fixed-size output vector, which is then processed by another recurrent network into an output. If the input is long, then the output vector would not be able to contain all relevant information, degrading the output. As evidence, reversing the input sentence improved seq2seq translation. The RNN search model introduced an attention mechanism to seq2seq for machine translation to solve the bottleneck problem (of the fixed-size output vector), allowing the model to process long-distance dependencies more easily. The name is because it "emulates searching through a source sentence during decoding a translation". The relative performances were compared between global (that of RNN search) and local (sliding window) attention model architectures for machine translation, finding that mixed attention had higher quality than global attention, while local attention reduced translation time. In 2016, Google Translate was revamped to Google Neural Machine Translation, which replaced the previous model based on statistical machine translation. The new model was a seq2seq model where the encoder and the decoder were both 8 layers of bidirectional LSTM. It took nine months to develop, and it outperformed the statistical approach, which took ten years to develop.